

SCHOOL OF ENGINEERING, CUSAT
Second Series Examination July 2024
Semester : IV
MECHANICAL ENGINEERING
B19-205-0402:METROLOGY & INSTRUMENTATION

Max. Marks: 50

Time: 2 Hours

Learning Outcomes:

Understand the concepts of measurements, limits, fits and tolerance.

Learn about measurement techniques of angles, tapers, and surface finish

Learn about applications of measuring instruments

Understand the measurement techniques for strain, force, torque, temperature, air pollution and nuclear radiation.

Bloom's Taxonomy Levels

Remember, L2- Understand, L3-Apply, L4-Analyse, L5- Evaluate, L6- Create)

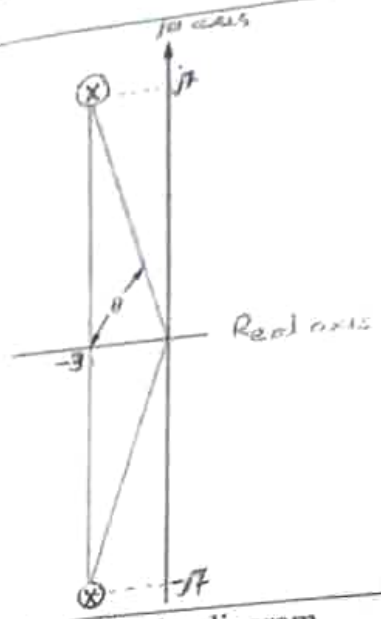
Programme Outcome

Part A		4x5= 20 Marks			
Answer ALL questions					
No.	Questions	Marks	BL	CO	PO
a)	Differentiate between surface roughness and waviness?	5	L2	2	1,2
b)	Explain the working of hydraulic load cell.	5	L1	4	1,2
c)	State and explain laws of thermocouple.	5	L1	4	1
d)	Differentiate between linear and non-linear systems. Write down an equation describing a linear time invariant system and how it differs from a linear time variant system.	5	L1	3	1,2
		2X15 = 30 Marks			
Part B		Marks	BL	CO	PO
a)	Explain the dynamic characteristics of a measurement system and differentiate between amplitude response, frequency response and phase response of a system.	10	L2	3	1,2
b)	Assume that a particular temperature probe approximates first-order behavior in a particular application, that it has a time constant of 6s, and (i) that it is suddenly subjected to a temperature step from 25°C to 150°C. What temperature will be indicated 10s after the process has been initiated? (ii) that it is suddenly subjected to a temperature step from 150°C to 25°C. What temperature will be indicated 10s after the process has been initiated?	5	L3	3	1,2

OR

III	a)	Define the following terms in surface texture measurements:- (i) Primary Texture. (ii) Secondary Texture. (iii) Lay. (iv) Sampling Length.	8	L1	2	1,2
	b)	Describe the method of evaluating roughness using (i) Peak to valley high method. (ii) C.L.A. method.	7	L1	2	1,2
IV	a)	Explain the working principle of Pneumatic load cell with neat sketches.	8	L1	4	1,2
	b)	Explain taylor hobson talysurf with neat sketches.	7	L1	4	1,2
OR						
V	a)	Diffrentiate between bonded and unbonded strain gauges.	7	L1	4	1,2
	b)	Explain the working of resistance temperature detector (RTD).	8	L1	4	1,2

PART A		CO	BL	Marks
Qo	Question	CO1	L2	4
a	Differentiate between a microprocessor and a microcontroller	CO3	L4	3
b	The location of poles of a second order system is shown in figure. What is the transfer function of the system			
c	Represent the following Boolean algebra equation in a PLC logic ladder diagram $Z = A \cdot \bar{B} \cdot C + \bar{D} + E$	CO2	L3	3



PART B		CO	BL	Marks
Qo	Question	CO2	L3	10
	Reduce the given block diagram to a single block and find the overall transfer function of the system (either by block diagram reduction or by signal flow graph).			
		CO3	L4	10
	A unity negative feedback control system has the plant $G(s) = \frac{50}{s(s+10)}$. Determine the percentage overshoot, 2% settling time x , rise time, and peak time a unit step input	CO3	L4	10
	When the system shown below is subjected to a unit step input, the response is like the figure shown on right side. Determine the value of K and T	CO3	L4	10
5	A washing machine has a pump to pump in water, a heater to heat water and another pump-to-pump water out. Write the ladder program to achieve the following: The pump is switched on to fill water for 3 minutes. Then the water is heated till the required temperature is reached, i.e., till the temperature sensor produce signal. On pressing the switch for the second pump, it pumps out water for 2 minutes and stops	CO4	L5	10

SCHOOL OF ENGINEERING, CUSAT
Second Series Examination July 2024

Semester : IV

MECHANICAL ENGINEERING (B - Batch)

19-200-0401: COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS

Time: 2 Hours

Max. Marks: 50

Course Outcomes:

1. Transform a region to another region using conformal mapping
2. Evaluate real integrals using residue theorem
3. Form and solve partial differential equation
4. Determine solution of partial differential equation for vibrating string and heat conduction

BL- Bloom's Taxonomy Levels

(L1- Remember, L2- Understand, L3-Apply, L4-Analyse, L5- Evaluate, L6- Create)

PO- Programme Outcome

Part A

4x5= 20 Marks

Answer ALL questions

Q. No.	Questions	Marks	BL	CO	PO
I a)	Find the singularities of the function $f(z) = \frac{9z+i}{z^3+z}$ and hence find the corresponding residues.	5	L2	2	4
b)	Find the Laurent series expansion of $f(z) = \frac{1}{z}$ in the region $ z - i > 1$.	5	L5	2	3
c)	Find a complete integral of $(p + q)(z - xp - yq) = 1$.	5	L5	3	2
d)	Derive one-dimensional wave equation.	5	L1	4	1

Part B (3)

2x15 = 30 Marks

		Marks	BL	CO	PO
II	By integrating around a unit circle, evaluate $\int_0^{2\pi} \frac{1}{5 - 3 \sin \theta} d\theta$.	10	L5	2	3
OR					
III	State Cauchy's Residue Theorem and use it to evaluate $\int_C \frac{4 - 3z}{z^2 - z} dz$ where $C: z = 2$.	10	L5	2	3
IV a)	Solve: $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 3z$.	5	L3	3	3
b)	Solve: $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$.	5	L3	3	3

OR

- V a) Solve: $r + s - 6t = \cos(2x + y)$. 5 L3 3 3
- b) Find the partial differential equation of all spheres of fixed radius having their centers in the xy - plane. 5 L2 3 2

- VI By the method of separation of variables, determine the solution to the one-dimensional heat equation. 10 L1 4 1

OR

- ✓ VII By the method of separation of variables, determine the solution to the one-dimensional wave equation. 10 L1 4 1

School of Engineering

B.Tech Degree IV semester Examination in Mechanical Engineering, July 2024

19-205-0405: HYDRAULIC MACHINERY

Time: 2 Hours

Max. Marks: 65

Test: 2

(B Batch)

Part A

5x3= 15 marks

Answer ALL questions

Q. No.	Questions	Marks
I a)	Define and explain hydraulic efficiency, mechanical efficiency and overall efficiency of a turbine?	3
b)	Draw neat sketches of the Pelton turbine and Kaplan turbine?	3
c)	Define specific speed of a turbine. Derive an expression for the same?	3
d)	Draw an indicator diagram of a reciprocating pump, considering the effect of acceleration and friction in suction and delivery pipes?	3
e)	Draw a neat sketch and explain the principle and working of hydraulic ram	3

Part B

5x10= 50 marks

Q. No.	Questions	Marks
II. a)	Find the efficiency of a free jet striking normally on a series of flat plates mounted on a periphery of a wheel?	4
b)	A jet of water moving at 10 m/s impinges on a vane shaped to deflect the jet through 120° degree when stationary. The vane is moving at 5 m/s, find the angle of jet so that there is no shock at inlet. What is the absolute velocity of jet at exit in magnitude and direction and the work done per second per unit weight of water striking per second? Assume that the vane is smooth.	6

- III** a) Define the following: 4
- (i) Gross head (ii) Net head
 (iii) Hydraulic Efficiency (iv) Overall efficiency.
- b) A Pelton wheel is to be designed for the following specifications: 6
 Shaft power = 11770 kW; Head = 380 meters; Speed = 750 rpm;
 Overall efficiency = 85 %; Jet diameter is not to exceed
 one-sixth of the wheel diameter. Determine i) The wheel
 diameter, ii) Number of jets required and iii) Diameter of jet.
 Take, Coefficient of velocity = 0.985 and Speed ratio = 0.45
- IV** a) What is the basis of selection of a turbine at a particular place? 4
- b) A Kaplan turbine develops 24650 kW power at an average head 6
 of 40 meters. Assuming a speed ratio of 2, flow ratio of 0.6,
 diameter of the boss equal to 0.35 times the diameter of the
 runner and an overall efficiency of 90.2%. Calculate the
 diameter, speed and specific speed of the turbine.
- V** ✓ a) What is an air vessel? Describe the function of the air vessel for 3
 reciprocating pumps
- b) The diameter and stroke length of a single-acting reciprocating 7
 pump are 12 cm and 20 cm respectively. The lengths of suction
 and delivery pipes are 8m and 25m respectively and their
 diameters are 8 cm. If the pump is running at 40 rpm and suction
 and delivery heads are 4m and 14m respectively, find the
 pressure head in the cylinder:
- (a) at the beginning of the suction and delivery stroke,
 (b) in the middle of suction and delivery stroke, and
 (c) at the end of suction and delivery stroke.
- Take atmospheric pressure head = 10.30 meters of water and
 $f = 0.009$ for both pipes.
- VI** a) Explain with the help of neat sketch, the principle and working 5
 of the following hydraulic devices:
 (i) Hydraulic intensifier (ii) Hydraulic accumulator
- b) Explain with the help of neat sketch, the principle and working 5
 of the following hydraulic devices:
 (i) Hydraulic lift (ii) Hydraulic crane